

YASASVI NELLORE

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PROFESSIONAL SUMMARY

- Full-Stack Engineer specializing in Java 17/Spring Boot microservices and React/TypeScript front ends, with 5+ years building cloud-native, event-driven systems on AWS.
- Proficient in developing RESTful and GraphQL APIs, optimizing relational and NoSQL databases (PostgreSQL, MongoDB), and deploying scalable applications.
- Passionate about leveraging user feedback and Open-Source technologies to drive continuous product improvement and operational excellence.

PROFESSIONAL EXPERIENCE

Software Engineer

Dec 2024 - Present

Twisted X Global Brands, Texas, USA

- Integrated custom **React/Node.js** applications and built a **Python**-based forecasting model (**Prophet, scikit-learn**) to automate inventory workflows, cutting manual tracking by 40% and boosting system uptime to 99.9%.
- Prototyped **MCP (Model Context Protocol)** integration with **NetSuite ERP** and internal **React/Node.js** tools using **WebSockets** and **context-aware APIs**, demonstrating AI-powered task automation with **Claude Code** and multi-agent orchestration in **Claude Desktop**.
- Deployed a real-time **Python** anomaly detection using Isolation Forest & **DBSCAN**, streaming data through **Kafka** and processing with **Pandas** and **NumPy** to flag inventory and order inconsistencies and cut manual reconciliation by 30%.
- Implemented **RESTful** and **GraphQL APIs** for ERP and e-commerce systems, increased data accuracy and improved production schedules, combined supplier and logistics APIs with **Node.js**, **MongoDB**, and **Oracle** for real-time data sync.
- Revamped **React.js** front end by introducing code-splitting and lazy loading in Webpack, other performance enhancing strategies, slashing page-load times from 3 s to 1.8 s.
- Optimized **Docker** and **Kubernetes** for containerization and scaling, reducing infrastructure costs and reduced system errors by 35% through automated testing with **Postman** and **Jest**.

Software Engineer

Dec 2023 - Dec 2024

JP Morgan Chase, Texas, USA

- Architected scalable **Spring Boot microservices** containerized with **Docker** on **AWS EKS** handling 5M+ events/day with zero downtime and led enterprise-grade API design using domain-driven design and **OpenAPI** standards.
- Developed and reviewed **Java 17 (Spring Boot)** and **React TypeScript** code with **OWASP** checks and **SonarQube** quality gates, catching and fixing 30+ critical defects before production.
- Automated remediation workflows using **Spring Boot** health checks and **Kubernetes** operators, eliminating 80% of recurring incidents and boosting system uptime to 99.99%.
- Provisioned and tuned **Elasticsearch/OpenSearch** clusters (**AWS Elasticsearch Service**), and **MongoDB** replica sets to index 50M+ documents improving full-text query performance by 70% under peak load.
- Streamed 1M+ document ingestion events/day via **Apache Kafka** and **Spark** Structured Streaming; orchestrated AI enrichment tasks with **Kubernetes** Jobs, ensuring lossless processing and ≤ 200 ms end-to-end resolution.
- Introduced circuit-breaker patterns, API Gateway rate-limiting, and **JWT/OAuth2** guards across **microservices**, achieved 99.99% uptime with proactive alerting via **Prometheus** and **Grafana**.
- Engineered a **MongoDB NoSQL** schema with compound indexes and aggregation pipelines, cutting average query time from 200 ms to 120 ms and supporting 5 $\times$ greater read throughput compared to the legacy SQL system.
- Spearheaded implementation of **Test-Driven Development (TDD)** and **Agile methodologies**, leading to the timely delivery of 95% of project milestones and a 15% reduction in post-deployment defects.

Java Developer

Jan 2021 - May 2022

Red Cloud Technologies, Hyderabad, India

- Delivered end-to-end **Java/J2EE** solutions on **Oracle** and **MySQL** within **Agile SCRUM** teams, leading requirement analysis, design, and deployment to reduce turnaround by 15% and boost CPU efficiency by 20%.
- Strengthened system throughput and resilience by architecting **Spring MVC microservices** with multithreading and hash-based caching layers, reducing average API response time from 200 ms to 160 ms under 1,000 RPS peak load.
- Deployed end-to-end **ETL** pipelines processing over 500 GB/day using **Apache Spark** for batch jobs, **Apache Kafka** for streaming ingestion, and **HDFS** for storage cutting job latency by 30% and delivering 99.9% reliability.
- Developed comprehensive API-driven front-ends, **SOAP** building **RESTful Spring Boot** and services consumed by an **React.js** UI, optimized client-side rendering and minimized page-load times by 25%.
- Assessed critical performance indicators using **AWS CloudWatch** and implemented optimizations that slashed server latency from 1.2 seconds down to an impressive 800 milliseconds under heavy traffic conditions.

- Engineered 10+ interactive internal dashboards using **React.js**, **D3.js**, and **Bootstrap 4** visualizing key KPIs (occupancy rates, lease renewals) for 200+ property managers and cutting manual reporting effort by 60%.
- Authored and maintained **RESTful APIs in Node.js** (v12) and **Express.js** (v4) with **MySQL** backends to aggregate lease, maintenance, and billing data boosting data-retrieval speed by 40% for dashboard queries.
- Configured client-side routing with **React Router** and managed complex application state via **Redux** implementing role-based views and data caching to elevate UX consistency and reduce load-time variability.
- Spearheaded test automation and CI by writing 100+ **Jest** unit tests and establishing **Travis CI/GitHub** pipelines achieving 85% code coverage, catching 75% of regressions pre-merge, and accelerating releases.

RESEARCH EXPERIENCE

Wireless Network Research & AI-Driven Data Processing

Jun 2023 – Dec 2023

Role: Graduate Research Assistant, Texas, USA

- Conducted performance analysis of Digital Twin models for **wireless** networks, identifying optimization strategies that improved simulated network efficiency by 28%.
- Fine-tuned **LLMs** using **NLP** techniques to automate insight extraction and improve predictive accuracy by 15% on wireless communication datasets.
- Programmed scalable data pipelines in **Python** for processing large-scale network data, leveraging **Pandas**, **NumPy**, and **Scikit-learn** to enhance feature engineering and reduce processing time by 35%.
- Leveraged **ML algorithms** and deep learning models with **TensorFlow** and **PyTorch** to forecast network load and traffic patterns, achieving 92% model accuracy.
- Utilized **Jupyter** Notebooks and deployed models using **AWS SageMaker** and **GCP Vertex AI**, accelerating training workflows and reducing iteration time by 40%.

EDUCATION

Master of Science in Computer Science

Aug 2022 - May 2024

University of Texas at Arlington, Texas, USA

Big Data & Cloud Computing, Machine Learning, Design Analysis & Algorithms

TECHNICAL SKILLS

- Languages:** Java, Python, JavaScript, Typescript, C++.
- Cloud & DevOps:** AWS (EC2, RDS, IAM), Azure, Google Cloud, Docker, Kubernetes, Terraform, Jenkins, Maven.
- Machine Learning & AI:** Supervised & Unsupervised Learning, Transformer-based models (BERT), Diffusion Models, Time Series Forecasting (Prophet), Topic Modeling, GPT-based LLMs
- Databases:** PostgreSQL, MySQL, MongoDB, Oracle 11g.
- Frameworks:** Spring Boot, Hibernate, Django, React.js, Express.js, Next.js, Node.js, Vue.js, Angular.
- Big Data:** Apache Spark, Hadoop, Hive, Pig, Scala, Kafka.
- Security & Authentication:** OAuth, OpenID, JWT.
- Tools & Testing:** Git, JUnit, Selenium, Postman, TDD, Jest.
- Design Patterns:** Singleton, Factory, Observer, MVC, Microservices, API design.

PROJECTS

- Chess Bellator** | Python-Machine learning algorithms- PyTorch-NumPy [GitHub](#)
 - A reinforcement-learning chess engine built in Python using PyTorch and NumPy to analyze board positions and iteratively improve its play.
- Automatic Job applications** | Python-OpenAI GPT-Selenium-OpenCV-Pandas-Docker [GitHub](#)
 - A Python automation tool that uses OpenAI GPT for cover-letter generation, Selenium for form submission, and OpenCV/Pandas for data extraction, all containerized with Docker
- Diffusion model to generate faces** | Huggingface- UNet2D- MetFaces [GitHub](#)
 - An image-generation pipeline leveraging Hugging Face’s UNet2D architecture trained on MetFaces to synthesize high-quality facial images.
- Diet AI Agent** | Python-Node.js-Express.js-LangChain-Postgressql [GitHub](#)
 - Full-stack Diet AI Agent that analyzes user data to generate personalized meal plans and track nutrition goals.

CERTIFICATIONS

AWS Solution Architect Associate - AWS.